

RF Line-Up

On-site G2

June 30, 2026

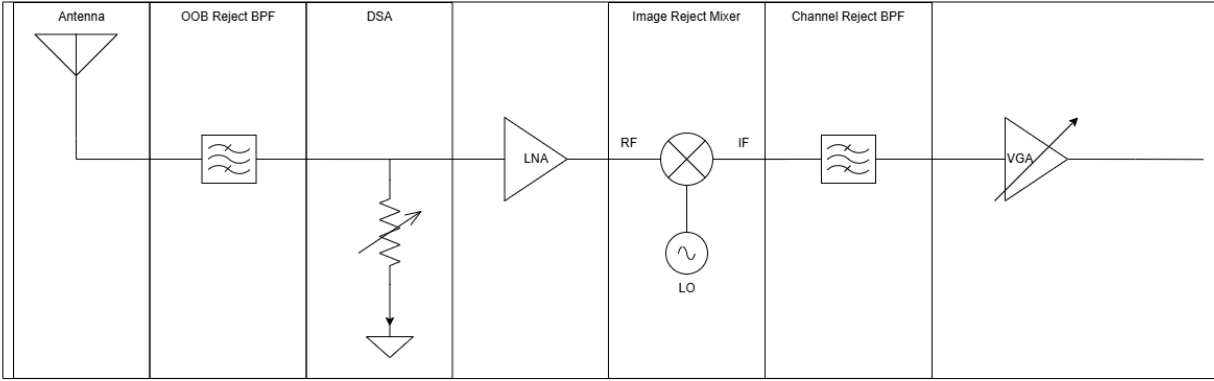


Figure 1: RF-LineUp Architecture

System Description Component Name	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
	Filter	Attenuator	LNA	Mixer	Filter
Gain (dB)	-2.4	0	24	12	-1.82
NF (dB)	2.4	2	2.4	1.8	1.82
P1dB (dBm)	100	-20	16	5	100
IP3 (dBm)	100	0	28	6	100

System Description Component Name	Stage 6
	VGA
Gain (dB)	10
NF (dB)	7.9
P1dB (dBm)	-23.3
IP3 (dBm)	-6

OUTPUTS

Parameter	Output
Gain (dB)	41.78
NF (dB)	6.07
P1dB (dBm)	-23.30
IP3 (dBm)	-6.04

Figure 2: Qorvo Calculator Result

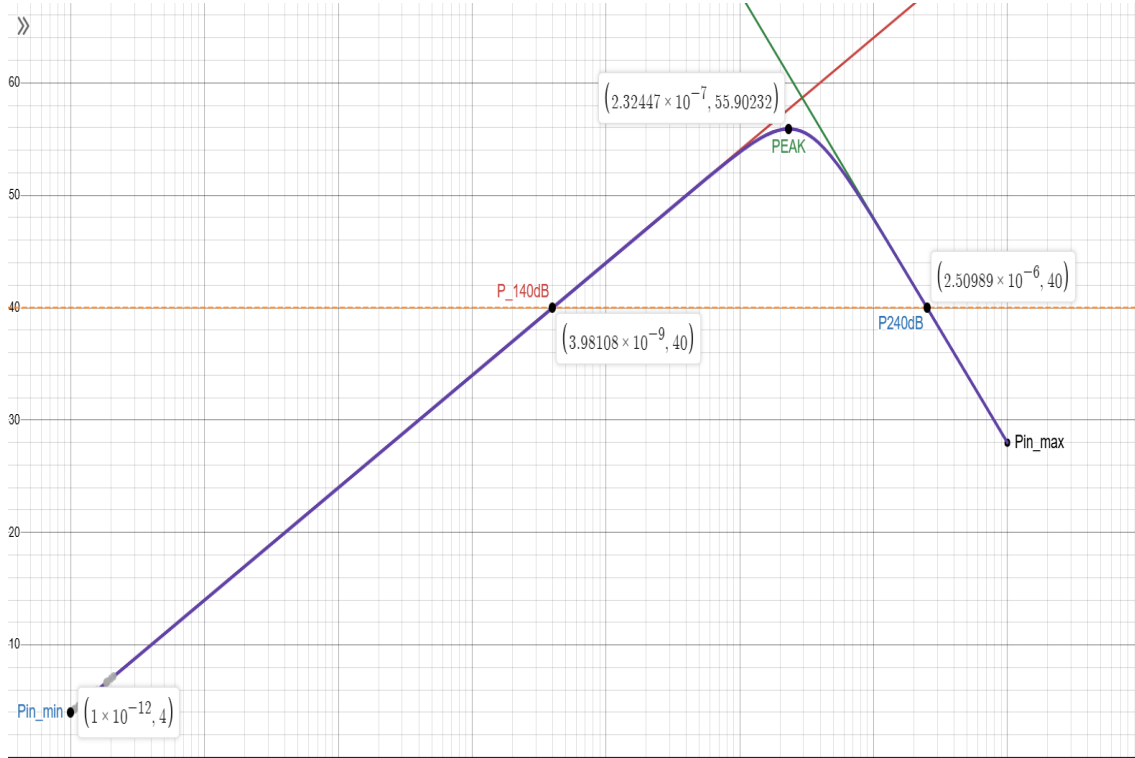


Figure 3: SFDR vs PIN.

As we can see here, at the lowest input power, the SNDR is 4 dB, which is close to the BPSK 10^{-3} BER SNDR requirement. Regarding the dynamic range over which the SNDR is higher than 40 dB—as required by 1024-QAM—we can see that this performance band starts at -54 dBm and ends at -26 dBm. Finally, the SNDR at the highest input power drops to 28 dB, reflecting the onset of non-linear distortion and gain compression in the system.

Please note the above figure was plotted assuming a constant NF of 6dB which is not true as we are using a VGA but for our purposes the NF in the higher power signal can be approximated as the NF at lower power without degrading the curve as the SNDR is dominated by distortion not noise, and the actual response is better in higher input power due to the VGA as shown in figure 4 on page 3.

Data / Results

Table 1: Component Specifications

Comp	Serial	Min Freq	Max Freq	Gain (dB)	NF(dB)	IP3 (dBm)	Price	Power
OOB BPF	MFBP-00082CH	13.8GHz	16.2GHz	-2.4	2.4	100	\$83	–
DSA	ADRF5700	0.1GHz	22GHz	(0 to -46)	2	1	\$54	8 mW
LNA	ADL5724	12.7GHz	15.4GHz	23.7	2.4	28	\$10	0.33 W
Mixer	HMC1113	10GHz (RF), 0(IF)	16GHz, 3.5	12	1.8	6	\$53	–
Channel Select BPF	RBP-75+	60MHZ	90MHZ	-2	2	100	1.857\$	0.1W
VGA	AD8372	5MHZ	85MHZ	-9to32	7.8	35	9.72\$	1.06W

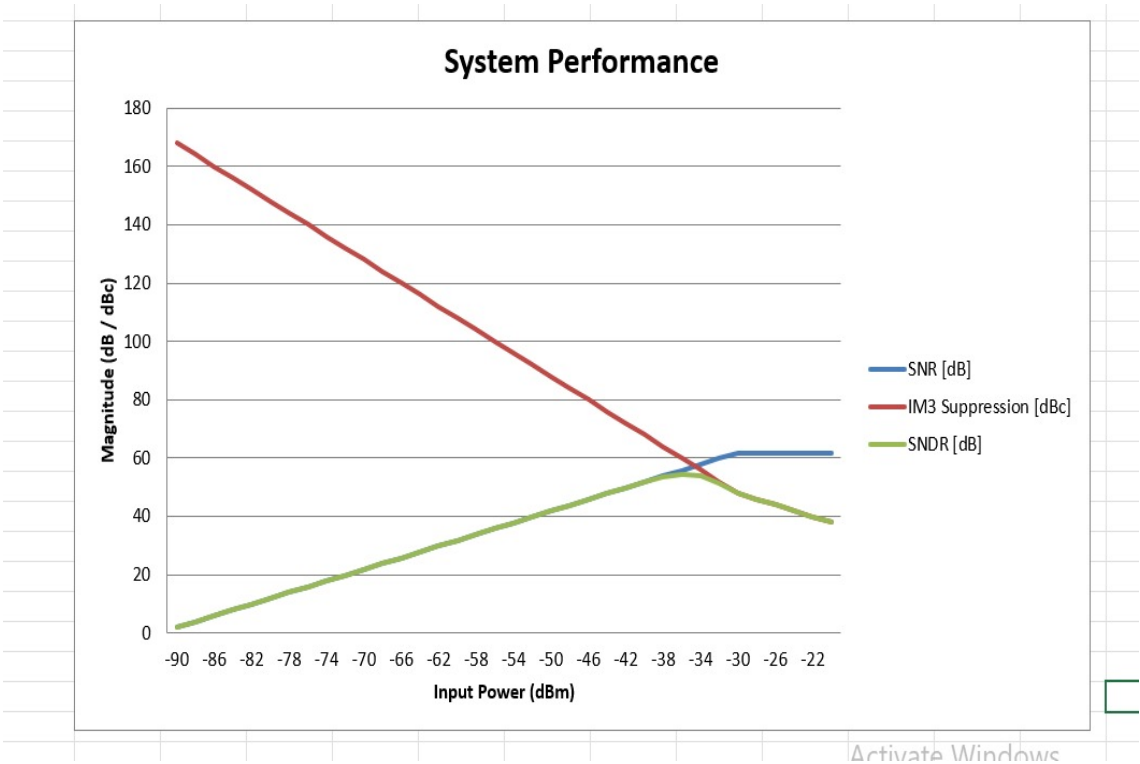


Figure 4: SFDR vs PIN with VGA Gain Sweep.